# Housing Discrimination and Algorithmic Bias

Submitted By: Group 2

Course: DATA-6550

# Introduction

Housing discrimination has been one of the significant factors in creating social and economic inequalities in society, particularly in the United States. Many discriminatory practices, such as racial covenants, redlining, and zoning laws, were common in housing markets in the United States in the past. Although these practices were outlawed decades ago, their impact can be seen in today's housing market through differences in property value, development, and access to financial opportunities.

In today's housing market, mortgage decisions are being made with the help of data-driven models, which are based on algorithms to support decision-making. Although these systems are unbiased, they are based on historical data, which already contains bias in today's context.

In this project, we will be looking into how historical housing discrimination affects today's mortgage lending market in the United States, with special consideration given to how data bias and algorithmic bias can affect decision-making in mortgage lending. The objective is to highlight how data science ethics play a role in decision-making in today's world, particularly in relation to housing market issues.

## Project Overview

This project examines the relationship between historical housing discrimination and modern lending patterns, focusing on two distinct types of bias:

1. **Data Bias**: How historical discrimination is embedded in modern data
   - Property values in areas with/without historical covenants
   - Neighborhood demographic and economic characteristics
   - Patterns of generational wealth and investment
2. **Algorithmic Bias**: How modern lending systems might perpetuate discrimination
   - Mortgage approval rates and terms
   - Risk assessment criteria
   - The use of potentially biased proxy variables

# Dataset Overview and Basic EDA

The dataset contains of 100,931 rows and 14 columns. It is a mortgage and housing dataset that combines lending information, applicant information, neighborhood characteristics and covenant related historical data. Initially before using this csv file in the jupyter file we have change the df = pd.read_csv("mortgage_covenant_data.csv") df.to_csv("hennepin_analysis.csv", index=False) for the easy access and understanding .

In the dataset the main columns were :
('activity_year,census_tract,derived_race,action_taken,loan_amount,property_value ,income,interest_rate,tract_minority_population_percent,tract_to_msa_income_percentage ,denial_reason-1,covenant_count,was_approved,covenant_density')

Some of the rows have missing values in it:
covenant_density → **89,762** missing
interest_rate → **22,806** missing
property_value → **17,440** missing
income → **12,731** missing
census_tract → **1,016** missing

## Key categories in the dataset

### Race distribution

Largest groups are:

- **White**: 63,627
- **Race Not Available**: 22,811
- **Asian**: 5,905
- **Black or African American**: 5,199
- **Joint**: 2,427

### Action taken

Main loan outcomes:

- 1 → 65,329
- 6 → 15,687

- 4 → 10,508
- 3 → 5,796
- 2 → 1,756

### Approval status

- Approved (True): 67,085
- Not approved (False): 33,846

- The data set also contains 791 duplicate rows in it.

The dataset used in the project has already been processed from various sources, which provide both historical and present-day housing-related data. The data includes housing-related data at the census tract level, which allows us to analyze the data at the neighborhood level.

The dataset includes the following key components:

· Covenant Density: It provides data on the presence of historical racial covenants, which shows the density of housing discrimination.

· Property Values: It provides data on the property values, which show the economic conditions of the neighborhood.

· Mortgage Lending Data: It provides data on the present-day mortgage lending, which shows the present-day decision-making process.

· Demographic Information: It provides data based on the census, which shows the demographic conditions of the neighborhood.

The data used in the project has already been processed from sources such as the Mapping Prejudice Project, the United States Census, the HMDA mortgage data, and more.

All these variables, put together, enable us to assess the possible effect that historical discrimination and its current presence in housing may have. For example, there could be a possible difference in property value or loan approvals in areas with a high incidence of historical covenants and those without this history.

The data has been clean and organized in a way that allows consistency across all variables. This makes it easier to compare the data. The data provides the basis upon which inequality will be analyzed, as well as the potential biases.

# Findings

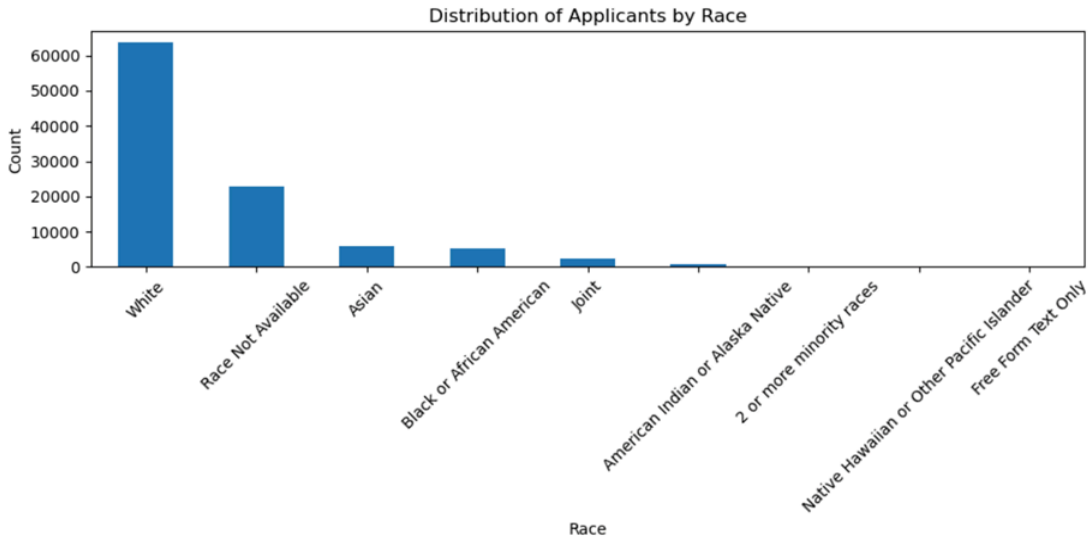
The analysis shows clear patterns of inequality:

- Property values increase with higher covenant density.
- Covenanted areas have higher median income and property values.
- Mortgage approval rates are higher in covenanted areas compared to non-covenanted areas.
- Approval rates vary significantly across racial groups, with some minority groups having lower approval rates.
- Income and property value show a positive relationship, influencing lending decisions.

These patterns suggest that historical advantages continue to impact modern economic outcomes.

# Data Bias

- Distribution of Applicants by Race




This is a bar chart showing how mortgage applicants are distributed according to their racial categories. The x-axis represents the race, and the y-axis represents the number of applicants.

From this graph, it is clear that white applicants are the largest number in this data set. They are way ahead of all other groups in terms of numbers. The second largest number is for those who are categorized as ‘Race Not Available.’ This could imply that most of this data set is incomplete in some way.

Among the other racial groups, Asian and Black or African American are present in moderate numbers, but nowhere near as high as white applicants. Other groups such as Joint, American Indian or Alaska Native, Native Hawaiian or Other Pacific Islander, and those who identify as multiple races are also present in small numbers.

Insight into the Data Set:

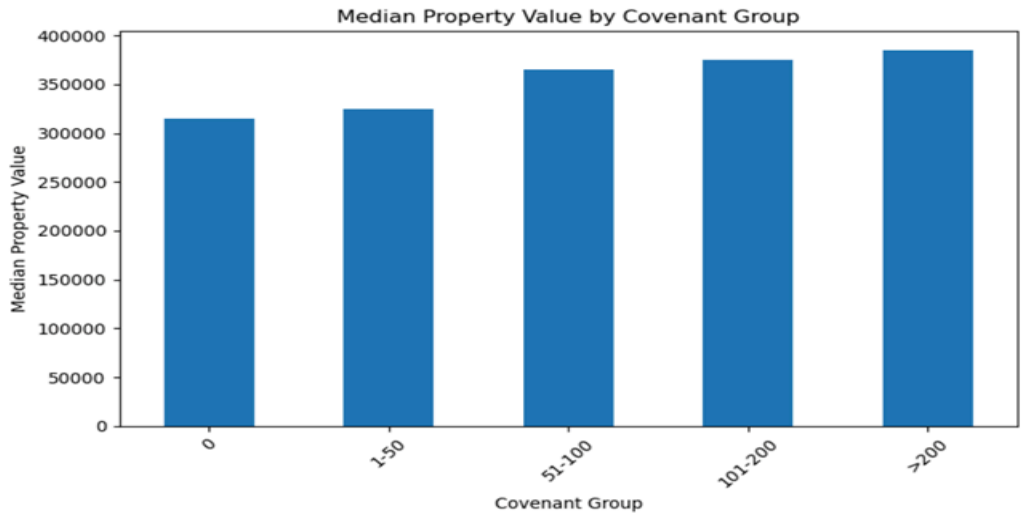
There is a huge imbalance in this data set. The white applicants are way ahead of all other groups. Some minority groups are also underrepresented in this data set. The large number of ‘Race Not Available’ could also imply that this data set is incomplete in some way.

Ethical Implication:

This data set is also relevant because it could imply that this data set could be biased in some way. If certain groups are underrepresented in this data set, then this could also imply that certain groups could be at a disadvantage in front of others.

- Median Property Value by Covenant Group

```
Median property value by covenant group:
covenant_group
0          315000.0
1-50       325000.0
51-100     365000.0
101-200    375000.0
>200       385000.0
Name: property_value, dtype: float64
```




This bar chart shows the median property value over different covenant density groups. The x-axis represents the different covenant density groups (0, 1-50, 51-100, 101-200, and over 200 covenants per area), and the y-axis represents the median property value in dollars.

What stands out:

- The median property value increases as the density of covenants increases.

- The areas without covenants have the least median property value of about $315,000.

- The median property value increases steadily as the density of covenants increases.

- The areas that have over 200 covenants have the highest median property value of about $385,000.

Interpretation:

From this chart, it is clear that areas that have many historical racial covenants have high property values. The covenants were used to bar non-white individuals from living in those areas. Therefore, this shows that areas that were privileged have been able to accumulate more wealth and resources over time.

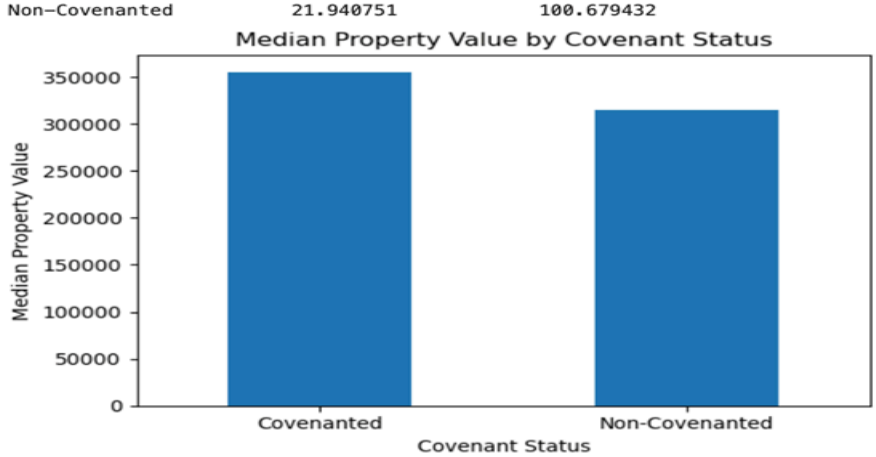
Ethical note:

This is an example of data bias that is brought about by past discrimination. The racial covenants may have been abolished in the past, but their effects can still be seen today. The effects of this bias may be felt if this information is used as input to algorithmic decisions.

- Median Property

```
Property / neighborhood summary by covenant status:
                 median_property_value  mean_property_value  median_income  \
covenant_binary
Covenanted                    355000.0        487349.429698          111.0
Non-Covenanted                315000.0        387560.383214           94.0

                 avg_minority_pct  avg_tract_msa_income
covenant_binary
Covenanted              30.374508            110.928425
Non-Covenanted          21.940751            100.679432
```




Valueby Covenant Status

This bar chart illustrates the differences in the median value of properties in covenanted and non-covenanted neighborhoods. The x-axis represents the covenanted and non-covenanted neighborhoods, and the median value of properties is represented on the y-axis.

Some key takeaways from this chart are: in covenanted neighborhoods, the median property value is around $355,000, while in non-covenanted ones, it is around $315,000. The difference in medians between these two types of neighborhoods is quite noticeable.

Additional data that support the differences in covenanted and non-covenanted neighborhoods are as follows:The mean value of properties in covenanted neighborhoods is higher than in non-covenanted neighborhoods, at around $487K and $387K, respectively.

- The median income levels are higher in covenanted neighborhoods than in non-covenanted neighborhoods, at 111 and 94, respectively.

- The average percentage of minorities is higher in covenanted neighborhoods than in non-covenanted neighborhoods, at 30 and 25, respectively.

Interpretation of the chart:

The chart suggests that neighborhoods that had a history of racial covenants have higher median values and income levels today. This suggests that these neighborhoods have enjoyed a higher standard of living and greater economic opportunities than non-covenanted neighborhoods.

Ethical implication of the chart:-This suggests how discrimination based on race in the past has had a lasting impact on the present.

From the algorithmic point of view, this suggests how a model based on the median value of properties and income levels may be biased in favor of covenanted neighborhoods, thus perpetuating discrimination in the present.

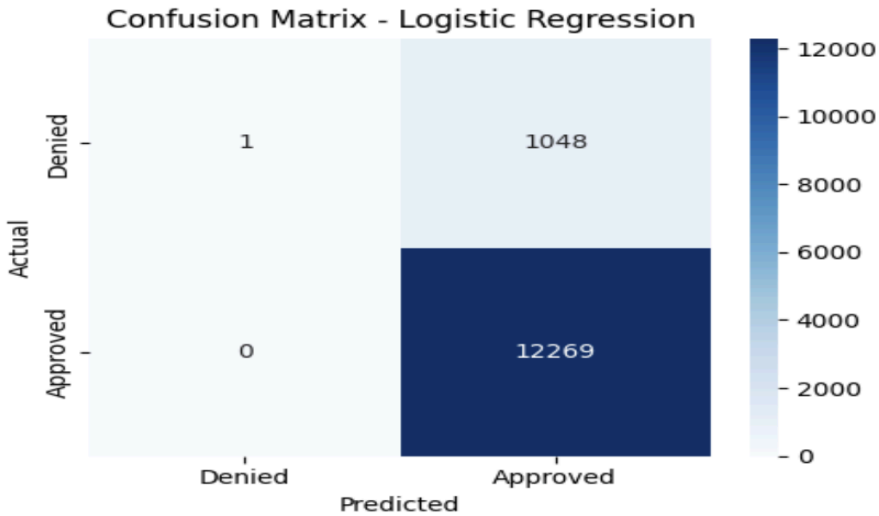
## Algorithmic Bias Analysis

### 1. Train Logistic Regression Model

```
=== Classification Report ===
              precision    recall  f1-score   support

      Denied       1.00      0.00      0.00      1049
    Approved       0.92      1.00      0.96     12269

    accuracy                           0.92     13318
   macro avg       0.96      0.50      0.48     13318
weighted avg       0.93      0.92      0.88     13318
```




This model uses Logistic Regression to predict whether a mortgage application will be approved or denied. The model is trained using income, property value, and other financial and neighborhood-related variables. The output of the model is a probability, which is then used to classify each application as either Approved or Denied.

## Key Observations:

- **High overall accuracy (92%)**
  The model is able to handle most of the cases, which is a clear indicator of the overall performance of the model.
- **Strong performance for approved cases**
  When it comes to approved cases, the model has a recall of 1.00. This implies that the model is able to handle all the approved cases.
- **Poor performance for denied cases**
  When it comes to cases that are denied, the model's performance is poor. The model has a recall of 0.00 for cases that are denied.
- **Confusion matrix shows misclassification**
  The confusion matrix shows that most of the cases that are denied are classified as approved, while most of the cases that are approved are correctly classified.
- **Class imbalance in the data**
  There is a huge difference between the number of approved cases and the number of denied cases.

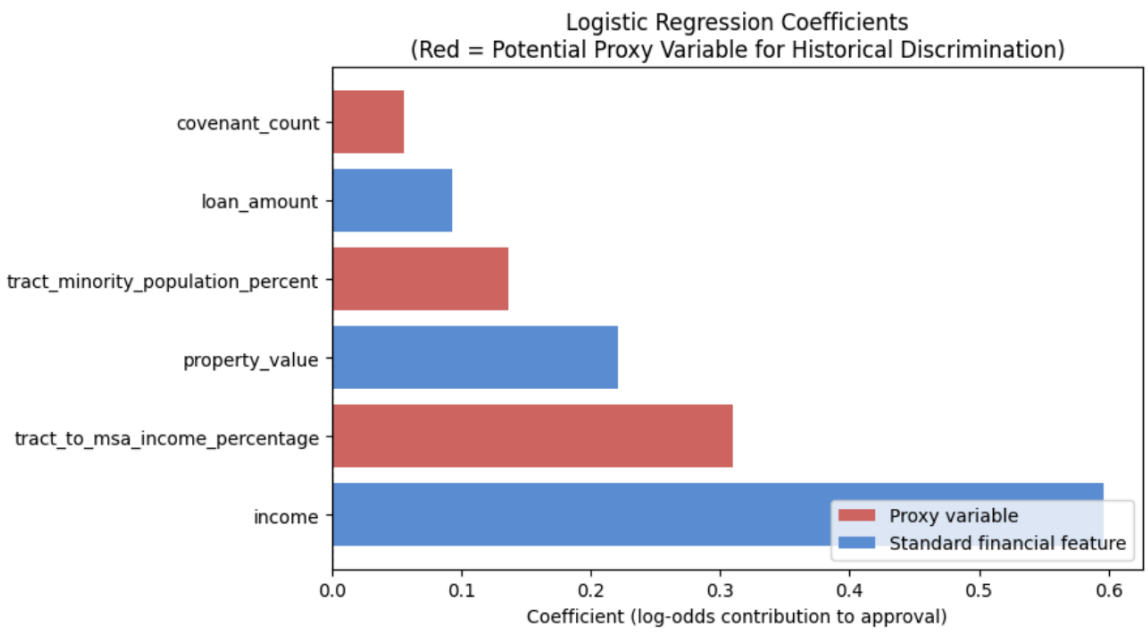
## Interpretation:

The model is heavily biased towards making predictions based on approved outcomes since it is trained on the majority class data. The model may have a high accuracy rate but is not effective in distinguishing between the denied cases.

## Ethical Implication:

There is a clear bias in the algorithm since the model is not treating the classes equally. The model is favoring the approvals and ignoring the denials, which may eventually lead to a faulty decision in a real-world scenario.

### 2. Feature Importance – Proxy Variable Analysis




```
Coefficient Table:
                          feature  coefficient  is_proxy
                           income     0.596755     False
   tract_to_msa_income_percentage     0.309216      True
                     property_value     0.220745     False
tract_minority_population_percent     0.136440      True
                       loan_amount     0.092403     False
                     covenant_count     0.055872      True
```

This bar chart shows the coefficients of our model as obtained through the Logistic Regression algorithm. The coefficients show how each feature influences the odds of approval. The x-axis shows the coefficient values, and the y-axis shows the model's features. Blue is used to show standard financial variables, and red is used to show variables that may act as proxies for historical discrimination.

## Key Observations:

- **Income has the highest coefficient**

Income is seen as having the largest influence, and this is because it has the highest coefficient value, implying that it greatly increases the probability of approval.

- **Tract-to-MSA income percentage**
  Next in line are tract to MSA income percentage, implying that approval is more likely for people living in more affluent areas.
- **Property value**
  Another significant influence is seen in property value, implying that association with more valuable properties increases approval likelihood.
- **Loan amount**
  Lastly, loan amount also plays a role, albeit to a lesser extent than income and property value.
- **Several variables shown in red**
  Three red-flagged features, namely covenant count, tract minority population percentage, and tract to MSA income percentage, are significant because they are likely proxy features in this model.

### Interpretation:

The figure demonstrates that this model relies substantially on both financial and neighborhood-based features in determining loan approval decisions. Although financial features are common in loan models, certain neighborhood features are also closely related to discrimination that may have been historically implemented. For instance, covenant count is closely related to racial housing discrimination in the past, while minority residents and tract income levels are closely related to discrimination in wealth and opportunities. As such, this model may indirectly address issues in social and racial discrimination without directly using racial features in loan decisions.

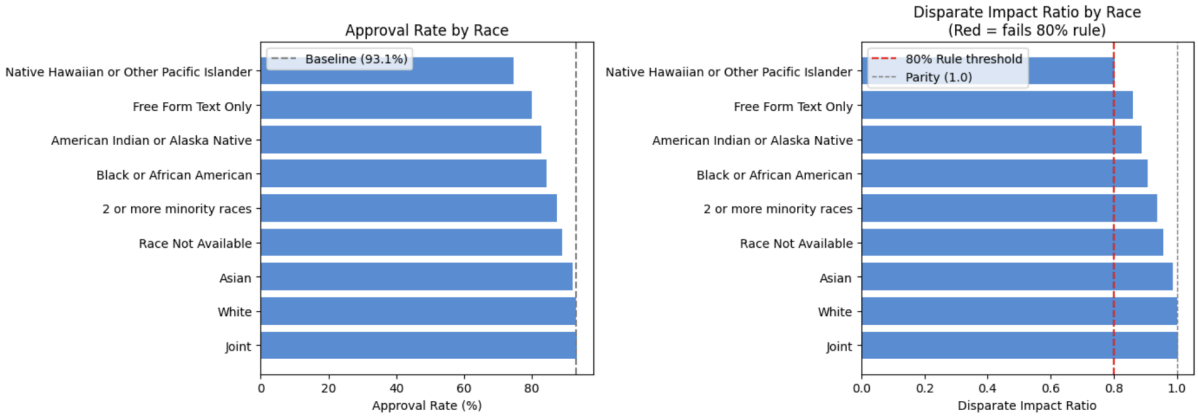
### Ethical Implication:

The features used in this model may also bring about issues in algorithmic bias. Although this model does not directly incorporate racial features in loan decisions, it may end up being biased in favor of certain groups and against others. As such, this model may end up favoring individuals from certain historically privileged groups while disadvantaging those from disadvantaged groups.

## 3. Disparate Impact Analysis

```
Baseline approval rate (White): 93.1%

Disparate Impact by Race:
                              derived_race  approval_rate_pct  disparate_impact_ratio  passes_80pct_rule
                                     Joint              93.35                  1.0028               True
                                     White              93.09                  1.0000               True
                                     Asian              91.88                  0.9870               True
                         Race Not Available              88.99                  0.9559               True
                    2 or more minority races              87.38                  0.9386               True
                   Black or African American              84.40                  0.9067               True
            American Indian or Alaska Native              82.75                  0.8889               True
                       Free Form Text Only              80.00                  0.8594               True
Native Hawaiian or Other Pacific Islander              74.71                  0.8026               True
```




We examine if there is any difference in the approval of mortgages for different races. We use the Disparate Impact Ratio to determine this. The baseline group is White applicants, and their approval rate is 93.1%.

## Key Points:

- The disparate impact ratio compares each group's approval rate to the baseline (White applicants).
- A value close to 1.0 indicates fairness (equal treatment), while lower values indicate potential disparity.
- Most racial groups have approval rates slightly lower than White applicants, for example:
    - Asian: ~91.9%
    - Black or African American: ~84.4%
    - American Indian or Alaska Native: ~82.7%
    - Native Hawaiian or Other Pacific Islander: ~74.7% (lowest)
- The second chart shows the 80% rule threshold (0.8), which is commonly used to detect discrimination.
    - If the ratio falls below 0.8, it may indicate unfair impact.
    - In this case, all groups pass the 80% rule, meaning no group falls below the threshold.

## Interpretation:

Although all groups technically pass the fairness threshold, there are still noticeable differences in approval rates across races.
Some minority groups consistently receive lower approval rates compared to White applicants.

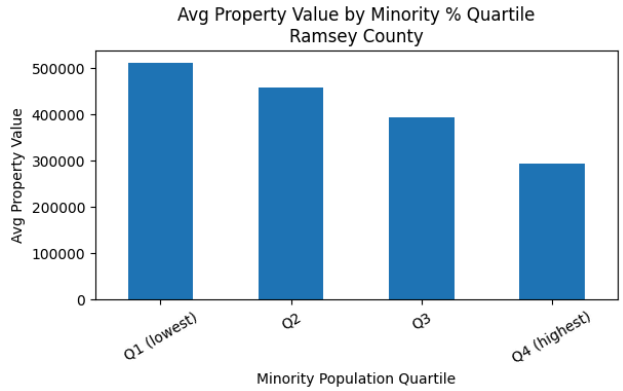
**Ethical Implication:**
The analysis shows that, while this system may technically be fair (as long as all groups are above the 80%), there are nuances to this, suggesting possible bias in how this algorithm was coded based on data sets.

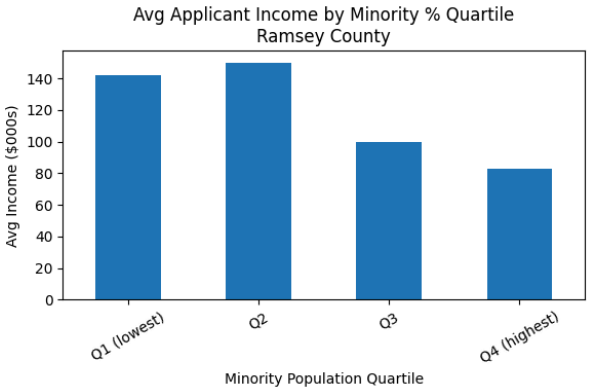
## 3.3 Ramsey County Extension

The research determined whether Hennepin County data patterns and algorithmic bias showed county-specific conditions or regional trends by studying Ramsey County which contains Saint Paul and Minnesota's second largest county. The CFPB Data Browser provided Home Mortgage Disclosure Act (HMDA) data for Ramsey County which includes FIPS code 27123 and contains 15,223 loan applications for 2024. The Mapping Prejudice Project only digitizes covenants in Hennepin County which creates the problem that covenant density data cannot be obtained for Ramsey; thus the study compares HMDA-derived lending indicators which both datasets share.
Note on year alignment: The Ramsey dataset covers 2024, while the Hennepin dataset reflects an earlier year. The comparison presents a limitation because it requires two different reference periods which researchers must consider when they study cross-county differences. Zero-to-one variations in absolute measurements between counties occurred because macroeconomic factors like increasing interest rates please read this explanation for additional details.

### 3.3.1 Data Bias in Ramsey County

Ramsey County shows the same economic opportunity decline which Hennepin County shows because of its minority neighborhood distribution. The lowest minority-population census tracts (Q1) display an average property value of $510,000 and an average applicant income of $141,000, while the highest-minority census tracts (Q4) show values of $299,000 and $82,000. The average property value has decreased by 41% while the average income has decreased by 42% across all quartiles, which reflects the ongoing effects of residential segregation and disinvestment in communities of color, which has persisted for decades.
The data bias present in the neighborhood-level differences arises from the mortgage lending system, which processes property valuations and income figures that contain historical inequality. A lending algorithm generates different outcomes based on race, even though it does not factor in race as a direct variable because it uses variables that have been historically shaped by racial factors.



### 3.3.2 Algorithmic Bias and Disparate Impact in Ramsey County

Table 1 displays mortgage approval rates together with disparate impact ratios which show racial differences between the two counties. The White approval rate in Ramsey County which reaches 85.5% acts as the essential baseline for EEOC 80% (four-fifths) rule. The major racial groups which have enough data available to study in Ramsey County all surpass 80% approval rates just like the Hennepin County groups do although the actual approval gaps between groups remain significant. Black or African American applicants in Ramsey County receive approval at a rate of 72.8% which results in a 12.7 percentage-point gap when compared to White applicants. American Indian or Alaska Native applicants experience an identical gap which reaches 73.1%.

Table 1. Mortgage Approval Rates and Disparate Impact Ratios by Race Hennepin vs. Ramsey County

| Race | Hennepin Approval % | Ramsey Approval % | Ramsey DI Ratio | Passes 80% Rule? |
|---|---|---|---|---|
| **White** | **92.1%** | **85.5%** | **1.000** | **Yes** |
| **Asian** | **90.6%** | **80.9%** | **0.946** | **Yes** |
| **Black or African American** | **84.6%** | **72.8%** | **0.852** | **Yes** |
| **American Indian/Alaska Native** | **81.9%** | **73.1%** | **0.855** | **Yes** |

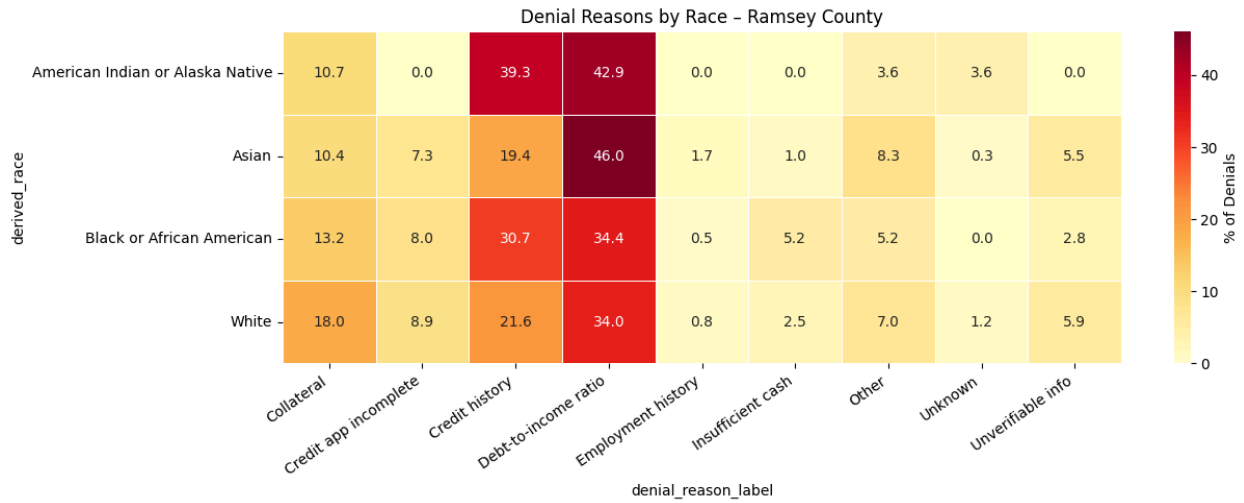
Ramsey County shows an approval rate of 82.8% which falls short of Hennepin County's 91.9% approval rate. The 9.1 percentage-point gap between the two counties remains constant through all racial groups because all groups in Ramsey face lower approval rates compared to their Hennepin County equivalents which indicates that differences in county-level lending institution types and application demographic trends and economic conditions create approval rate differences that extend beyond racial factors.

### 3.3.3 Denial Reasons by Race

The heatmap of denial reasons for Ramsey County shows that debt-to-income ratio (DTI) and credit history together account for all racial group denials which matches the findings from Hennepin. The two reasons show different importance levels between racial groups:

- For American Indian or Alaska Native applicants, collateral (39.3%) and debt-to-income ratio (42.9%) together account for over 80% of denials, the highest combined share of any group suggesting these applicants disproportionately face barriers related to both asset valuation and income adequacy.
- For Asian applicants, debt-to-income ratio is the single largest denial category at 46.0%, followed by credit history at 19.4%.
- For Black or African American applicants, DTI (34.4%) and credit history (30.7%) dominate, with a combined share of 65.1%.
- White applicants show a more distributed denial profile, with DTI (34.0%), collateral (18.0%), and credit history (21.6%) each playing a significant role.

The data patterns support the bias argument because DTI ratios and credit scores function as racial non-neutral variables. The economic circumstances of neighborhoods and credit access disparities and generational wealth disparities arise from discriminatory policies. The creditworthiness assessment methods used in Ramsey function as racial and historical disadvantage proxies just like they do in Hennepin.




### 3.3.4 Interest Rate Disparities

Among approved loans, Ramsey County borrowers pay statistically significantly higher interest rates than their Hennepin counterparts. A Welch two-sample t-test yields $t = -33.34$ ($p < 0.0001$), confirming that the difference is not due to chance. Table 2 shows median interest rates by race and county.

**Table 2. Median Interest Rates on Approved Loans by Race Hennepin vs. Ramsey County**

| Race | Hennepin Median Rate | Ramsey Median Rate | Ramsey Higher? |
|---|---|---|---|
| **White** | **6.750%** | **6.990%** | **+0.240 pp** |
| **Asian** | **6.625%** | **6.990%** | **+0.365 pp** |
| **Black or African American** | **6.625%** | **6.875%** | **+0.250 pp** |
| **American Indian/Alaska Native** | **6.875%** | **7.000%** | **+0.125 pp** |

Ramsey County borrowers receive higher interest rates than Hennepin County borrowers according to data from approved loans. The Welch two-sample t-test showed $t = -33.34$ ($p < 0.0001$), which demonstrated that the observed difference happened because of actual factors instead of random variability. Table 2 shows median interest rates by race and county.

All racial groups in Ramsey County face higher interest rates ranging from 0.125 to 0.365 percentage points when compared with their Hennepin County counterparts. Throughout both

counties, the racial hierarchy remains intact as White applicants receive the lowest median rates while American Indian or Alaska Native applicants face higher rates. The interest rate disparities found across multiple counties demonstrate that Hennepin County does not have exclusive interest rate gaps but rather shows a widespread pattern which exists throughout Twin Cities metropolitan area mortgage lending. A higher rate on an approved loan compounds over time because a 0.25 percentage-point difference on a $300,000 30-year mortgage will cost approximately $15,000 in total interest, which creates significant ongoing wealth transfer disadvantages for minority borrowers.

# 4. Discussion

## 4.1 Implications for Modern Lending

The analysis shows that existing racial discrimination practices which began in Hennepin and Ramsey counties have created two distinct modern mortgage assessment methods which track their historical home loan evaluation patterns. First, data bias occurs when property values and applicant incomes and neighborhood income indices show economic patterns which developed through racial covenants and redlining practices. The algorithm training process uses property value and income data from census tracts which have high racial covenant historical density, which establishes a system that maintains all past exclusion patterns.

Second, algorithmic bias occurs when a logistic regression model uses race-neutral input features to predict outcomes which show racial differences between groups. The disparate impact analysis confirms this practically: Black or African American and American Indian or Alaska Native applicants are approved at 84.6% and 81.9% of the White approval rate in Hennepin, and 85.2% and 85.5% in Ramsey. Even approved minority borrowers pay higher median interest rates, adding thousands of dollars in costs over the life of a loan.

The research results throughout the two counties demonstrate that systemic problems affect the entire region instead of targeting particular lenders or local areas.

## 4.2 Proposed Mitigation Strategies

The treatment of these biases requires action at different levels, as:

- Regular third-party algorithmic audits test lending models for disparate impact before and after deployment. The results of these tests must be reported to regulatory authorities.
- Alternative credit data which includes rent and utility and telecom payment records helps to decrease denial rates for applicants who have insufficient traditional credit histories..
- Valuation reform developing rating methods not rooted in historically segregated comparables is a clear starting point to openly and explicitly overcome data biasness and growth-molestations indicated therein.
- The program provides targeted down payment assistance to eligible candidates who reside in historically covenanted census tracts, which enables them to overcome the financial challenges caused by generational wealth gaps that result in collateral and cash-reserve denials.

- The government has introduced some amendments to the Community Reinvestment Act as per a portion of the auspicious bipartisanship agreement included in the financial-regulation-reform legislation package that had been composed in 1977 and required lending agencies to report on the underserviced and disinvested areas by establishing lending patterns at the Census tract.

The regional nature of disparities requires multiple organizations to work together for effective reform which needs to extend beyond county boundaries to include all lending institutions and appraisers and regulators and local government entities.

## 5. Limitations and Future Work

### 5.1 Limitations

- The Ramsey County HMDA data covers 2024 while the Hennepin dataset reflects an earlier year, which leads to cross-county differences in rates and approvals that might stem from macroeconomic conditions.
- The Mapping Prejudice Project provides covenant density data which only exists for Hennepin County. The Ramsey extension uses tract minority population percentage as its proxy which shows the result of discrimination but not how it actually happened.
- The logistic regression works as an understandable model which shows how proxy variable bias affects the results but does not function as a reverse-engineered audit of any actual lender model. Production underwriting systems use more complex models (e.g., gradient boosting) which may produce different results.
- The HMDA dataset does not include credit scores which serve as one of the strongest mortgage approval predictors. The analysis needs this control to achieve proper separation between neighborhood effects and personal financial elements.
- The analysis focuses on the Twin Cities metropolitan area and its results will not apply to markets which have different housing patterns and lender types.

### 5.2 Future Work

- The two county datasets need to be aligned toHMDA reporting year 2023 because this alignment will improve cross-county comparison results. The project needs to extend its covenant digitization work to Ramsey County so Saint Paul can implement Hennepin data bias analysis through direct data replication.
- The application of SHAP values enables researchers to use a gradient boosting model for identifying interaction effects which exist between proxy variables yet remain hidden to logistic regression analysis.
- The researchers will use HMDA LEI codes to separate the disparate impact analysis according to each lender which will help them determine whether gaps exist only in certain financial institutions.
- The researchers will assess current down payment assistance programs through matched-sample design which will evaluate whether specific financial support measures succeed in closing racial homeownership gaps.